

MARTIN LOUIS

AI-NATIVE ENTERPRISE PLATFORM & CUSTOMER EXPERIENCE ENGINEERING LEADER

Agentic Systems | Enterprise AI Architecture | Autonomous Servicing | Platform Strategy

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EXECUTIVE SUMMARY

AI-native enterprise platform leader with 20+ years building systems that connect real-time signals, semantic context, retrieval and LLMs, agents, enterprise tools, workflows, governance, and people to reliable customer and business outcomes. Set platform direction within a 30+ engineer organization across five teams, with a direct delivery portfolio of approximately 20 engineers. Scaled a two-engineer initiative to a ten-engineer platform team over 18 months, combining reference architecture, investment strategy, organizational design, platform economics, and multi-team execution.

Selected impact: 30+ engineers / 5-team organizational scope | 2 to 10 engineers in 18 months | 50M users served | 11 strategies associated with \$110M+ annual TPV | 200+ product-manager stakeholder population | 6 peer-reviewed papers | IEEE Senior Member (2025)

LEADERSHIP & ARCHITECTURE

AI-Native Enterprise Platforms - Agentic AI & Autonomous Workflows - Reference Architecture - Semantic Layers & Canonical Domain Events - Data Contracts & Event-Driven Systems (Kafka) - Retrieval, RAG & Knowledge Graphs - MCP-Based Tool Integration - Workflow State & Orchestration - Evaluation, Observability & Reliability - Governance, Guardrails & Human-in-the-Loop - Customer Experience & Operating Economics - Platform Adoption & Reuse - Engineering & Organizational Leadership

PROFESSIONAL EXPERIENCE

PayPal - Senior Engineering Manager, AI-Powered FinTech & Customer Success

2022 - Present

Austin, TX | Enterprise AI platforms, agentic systems, and autonomous customer servicing

- Set platform vision and north-star reference architecture within a 30+ engineer organization across five teams, with an approximately 20-engineer direct delivery portfolio spanning knowledge and answer platforms, proactive servicing, intelligent routing, predictive AI, and autonomous agents.
- Lead a reusable enterprise agentic AI framework combining grounded LLM/RAG patterns, multi-agent and multi-turn orchestration, MCP-based tool interfaces, workflow state, evaluation, observability, guardrails, and human-in-the-loop execution.
- Architected a shared semantic intelligence layer using canonical domain events, data contracts, behavioral and transactional signals, a product knowledge graph, retrieval, ranking, and decisioning across 20+ Kafka topics and 12 domains.
- Reimagined content and knowledge operations as AI-native workflows spanning structured models, generation, translation, publishing, review, legal approval, and human oversight.
- Scaled a two-engineer initiative to a ten-engineer platform team over 18 months. The cross-functional platform served 50M users and enabled 11 intervention strategies associated with more than \$110M in annual transaction-value recovery.
- Built the investment case for a \$16.5M multi-year program, aligned four partner organizations on shared architecture and governance, and supported a 200+ product-manager stakeholder population as adoption progressed toward a 70% target.
- Drive portfolio decisions through customer and operating economics, balancing experience signals, transaction recovery, reliability, cost-to-serve, TCO, adoption, reuse, answer quality, and delivery velocity while hiring and developing engineers and managers.

PayPal - Software Engineering Manager

2018 - 2022

Austin, TX | Personalization, real-time decisioning, and intelligent customer self-service

- Evolved enterprise content management into a reusable intelligence platform, connecting knowledge, behavioral signals, analytics, personalization, and experience delivery.
- Architected a real-time Java/Spring recommendation and personalization engine; improved recommendation accuracy approximately 25% and reduced search time approximately 40% for 10M+ monthly active users.
- Secured executive sponsorship and multi-million-dollar investment while leading architecture, delivery, talent development, and operational excellence.

PROFESSIONAL EXPERIENCE | CONTINUED

PayPal - Member of Technical Staff / Senior Software Engineer / Software Engineer 3

2011 - 2018

Austin, TX and Chennai, India | Search, knowledge, retrieval, and self-service foundations

- Designed large-scale search and knowledge capabilities for 50M+ users using Java, Lucene, Oracle Knowledge/Search, NLP, and relevance engineering across web, mobile, and conversational channels.
- Developed fuzzy-matching and search-relevance capabilities that reduced resolution time approximately 35%, building deep technical lineage in retrieval quality, ranking, semantic matching, and trusted knowledge systems.
- Built reusable search and knowledge services and automated AWS deployments across APIs, data integration, reliability, and production operations; earned multiple PayPal SPOT Awards.

Computer Sciences Corporation (CSC) - Associate Manager / Senior Software Engineer / Technical Lead

2008 - 2011

Chennai, India | Enterprise software delivery for energy, financial-services, and insurance clients

- Led enterprise software delivery across requirements, architecture, build, test, deployment coordination, and support for complex economic planning, forecasting, and reserve-estimation systems.
- Coordinated onsite and offshore teams, user acceptance testing, client-facing implementation planning, and technical delivery across Java, .NET/C#, and data-intensive workflows.

Covansys India - Senior Software Engineer / Associate Projects / Associate Trainee

2005 - 2008

Chennai, India | Enterprise web, integration, e-commerce, and innovation

- Built enterprise web applications, rich-internet interfaces, reusable UI components, database objects, and integration layers for retail, manufacturing, financial-services, and energy clients.
- Recognized for innovation, technical excellence, and delivery execution through Covansys PRIDE Recognition (2006 and 2007) and the CORE Excellence Award.

SELECTED PUBLICATIONS | 6 PEER-REVIEWED PAPERS, 2019-2023

- Scaling Trustworthy AI: A Framework for Responsible System Design - GJETA, 2023
- Using Large Language Models to Address the Cold-Start Problem in ML Training Data for E-Commerce Product Listing Generation - WJARR, 2023
- Transforming Commerce: The Role of NFTs in Enhancing Customer Engagement and Loyalty - IJSRA, 2022
- Personalized Recommendation Systems for Customer Self-Service and Promotions - WJARR, 2021
- Semantic Search Leveraging Product Knowledge Graphs for Financial Products in Web 2.0 Ecosystems - IRE Journals, 2020
- Multilingual Search Optimization Using BERT AI and Product Knowledge Bases - IRE Journals, 2019

RECOGNITION & THOUGHT LEADERSHIP

- IEEE Senior Member: elevated in 2025. Confirmed 2025 memberships included the Computer Society, Control Systems Society, Signal Processing Society, and Technology and Engineering Management Society.
- IEEE CCIC 2026 Industry Keynote: delivered AI as an Entrepreneurial Operating System: Reimagining How Small Businesses Are Built on February 7, 2026.
- Media: AI Journal (September 30, 2025 interview); Techstrong.ai (August 6, 2025 article); AllTech Magazine (August 4, 2025 profile); SiteProNews (June 11, 2025 article).
- Additional recognition: PayPal SPOT and performance awards; PayPal Austin Hackathon Bounty Winner (2018); Harmony Public Schools District Science Fair Judge (2020); CSC Chairman's Technical Excellence Award nominee.

EDUCATION & CERTIFICATIONS

- The University of Texas at Austin - Graduate-level coursework in the Post-Graduate Program in Cloud Computing (2022).
- Bharathiar University, India - Master of Computer Applications (2005) and B.Sc. Computer Science (2002).
- Certifications: Certified Product Manager, Product School (2020) - Six Sigma Green Belt - Sun Certified Java Programmer - IBM Certified Developer and Database Associate.